

Parametric Noise Coherence Matrix Estimation through Neural SNR Regression for MVDR Beamforming

Chunrui Zhao*, Yujie Zhu*, Jilu Jin[†], Xueqin Luo[†], Gongping Huang*, Jingdong Chen*, and Jacob Benesty[‡]

*Electronic Information School, Wuhan University, Wuhan, China

[†]CIAIC, Northwestern Polytechnical University, Xi'an, China

[‡]INRS-EMT, University of Quebec, Montreal, Canada

Abstract—Minimum Variance Distortionless Response (MVDR) beamforming is a widely adopted technique for multichannel speech enhancement. However, its performance is highly sensitive to the accurate estimation of the noise covariance matrix (NCM). Traditional data-driven methods often focus on estimating time-frequency masks or full covariance matrices, leading to high-dimensional learning tasks and increased computational overhead. In this paper, we introduce a model-driven framework that reformulates NCM estimation as a low-dimensional, frequency-wise signal-to-noise ratio (SNR) regression problem. By leveraging the parametric structure of a single-source far-field model, we show that the noise coherence matrix can be reconstructed from the observed signal coherence using only the input SNR. A lightweight neural network is used to estimate the frequency-dependent SNR, which is then incorporated into a closed-form coherence matrix reconstruction for MVDR beamforming. Experimental results demonstrate that the proposed method provides significant speech enhancement performance, robustness, and low computational complexity, making it a practical and efficient solution for real-world applications.

Index Terms—Microphone arrays, MVDR beamformer, SNR estimation, noise covariance matrix estimation.

I. INTRODUCTION

Microphone arrays, in conjunction with beamforming, play a crucial role in modern speech communication systems by utilizing spatial information to enhance speech quality in challenging acoustic environments [1–8].

Over the years, a wide range of beamforming algorithms have been developed, including delay-and-sum, filter-and-sum, superdirective, differential, adaptive, and neural beamformers [9–11]. Among these, the minimum variance distortionless response (MVDR) beamformer stands out due to its optimality in minimizing the output noise power while preserving the desired signal from a specific direction [12], [13]. This property has made MVDR an effective and widely adopted method for multichannel speech enhancement [14], [15]. A critical aspect of the MVDR beamforming formulation is the noise covariance matrix (NCM), which encapsulates the spatial characteristics of the noise field [16–18]. In real-world applications, where the NCM is typically unknown, a key challenge is reliably estimating it from the microphone array observations [19], [20]. Traditionally, the NCM has been estimated using statistical methods that assume signal stationarity [21],

[22]. More recently, data-driven approaches have become popular, employing techniques such as spatial clustering [23] or deep neural networks [24–26] to estimate time–frequency masks for constructing the NCM. However, these methods often involve estimating masks for every time-frequency bin, which can be computationally expensive and potentially redundant when building a robust NCM. This highlights the need for methods that leverage the parametric structure of the signal model, particularly under the single-source assumption.

In this work, we exploit the parametric structure of the signal model to reformulate the NCM estimation as a signal model-driven SNR regression problem. This approach is motivated by a fundamental theoretical relationship that characterizes the composition of the spatial covariance matrix: under the single-source far-field model, with a known steering vector, the coherence matrix of the observed signals can be expressed as a weighted sum of a rank-one speech coherence matrix and the NCM [1]. The weight for this combination is entirely determined by the input signal-to-noise ratio (SNR), or equivalently, the Wiener gain. This implies that, once the input SNR is estimated, the NCM can be constructed analytically in closed form, without the need for explicit time–frequency mask estimation. To estimate the input SNR, a lightweight neural network is employed, which then enables the direct computation of the NCM from the observed signal statistics. The key contributions of this work are: 1) the reformulation of NCM estimation as a low-dimensional, model-constrained SNR regression task; 2) the development of a lightweight neural network for input SNR estimation; and 3) the construction of a robust and computationally efficient MVDR beamforming system by integrating the analytically derived NCM. Experimental results demonstrate that the proposed method outperforms conventional approaches, including IMCRA-based SNR estimation baselines.

II. SIGNAL MODEL AND PROBLEM FORMULATION

Consider an array consisting of M omnidirectional microphones, with all microphone positions known *a priori*. Suppose a speech source of interest is located in the sound field and impinges on the array from the direction θ_s . The received signal in the short-time Fourier transform (STFT) domain at the m th microphone, $m = 1, 2, \dots, M$, is given by [2]

$$Y_m(f) = X_m(f) + V_m(f), \quad (1)$$

where $X_m(f)$ and $V_m(f)$ denote the desired speech and noise components, respectively. It is assumed that all signals have zero mean, and the speech components are uncorrelated with the noise components.

Stacking the observations from all microphones into a vector gives [28]

$$\mathbf{y}(f) = [Y_1(f) \ Y_2(f) \ \cdots \ Y_M(f)]^T \quad (2)$$

$$= \mathbf{x}(f) + \mathbf{v}(f), \quad (3)$$

where $\mathbf{y}(f)$, $\mathbf{x}(f)$, and $\mathbf{v}(f)$ denote, respectively, the noisy speech, clean speech, and noise vectors, and the superscript T represents the transpose operator.

In the special case of an anechoic environment with negligible attenuation effects, the speech component at the m th microphone is related to the reference source signal $X(f)$ through a phase shift [27], i.e.,

$$X_m(f) = e^{-j2\pi f \tau_m} X(f). \quad (4)$$

where τ_m denotes the time difference of arrival (TDOA) between the m th microphone and the reference point. Under this condition, the observation vector can be expressed as

$$\begin{aligned} \mathbf{y}(f) &= [e^{-j2\pi f \tau_1} \ e^{-j2\pi f \tau_2} \ \cdots \ e^{-j2\pi f \tau_M}]^T X(f) + \mathbf{v}(f) \\ &= \mathbf{d}_{\theta_s}(f) X(f) + \mathbf{v}(f), \end{aligned} \quad (5)$$

where $\mathbf{d}_{\theta_s}(f)$ is the source propagation vector. Then, the covariance matrix of the observation signals is [29]

$$\begin{aligned} \Phi_{\mathbf{y}}(f) &= E[\mathbf{y}(f)\mathbf{y}^H(f)] \\ &= \phi_X(f)\mathbf{d}_{\theta_s}(f)\mathbf{d}_{\theta_s}^H(f) + \Phi_{\mathbf{v}}(f), \end{aligned} \quad (6)$$

where $\phi_X(f) = E[|X(f)|^2]$ is the variance of the desired speech signal, $\Phi_{\mathbf{v}}(f) = E[\mathbf{v}(f)\mathbf{v}^H(f)]$ is the NCM, $E[\cdot]$ denotes mathematical expectation, and the superscript H is the conjugate-transpose operator.

Beamforming with microphone arrays is typically carried out in the frequency domain by applying a complex-valued filter $\mathbf{h}(f)$ of length M to the observation vector $\mathbf{y}(f)$ [30], i.e.,

$$\begin{aligned} Z(f) &= \mathbf{h}^H(f)\mathbf{y}(f) \\ &= \mathbf{h}^H(f)\mathbf{d}_{\theta_s}(f) X(f) + \mathbf{h}^H(f)\mathbf{v}(f), \end{aligned} \quad (7)$$

where $Z(f)$ is an estimate of $X(f)$.

In practice, it is desirable to impose a distortionless constraint on the beamformer [31], i.e.,

$$\mathbf{h}^H(f)\mathbf{d}_{\theta_s}(f) = 1, \quad (8)$$

so that the source signal arriving from θ_s passes through the beamformer without distortion.

The MVDR beamformer is obtained by minimizing the output noise power subject to the distortionless constraint, and is given by [31]

$$\mathbf{h}_{\text{MVDR}}(f) = \frac{\Phi_{\mathbf{v}}^{-1}(f)\mathbf{d}_{\theta_s}(f)}{\mathbf{d}_{\theta_s}^H(f)\Phi_{\mathbf{v}}^{-1}(f)\mathbf{d}_{\theta_s}(f)}. \quad (9)$$

The implementation of the MVDR beamformer requires knowledge of the noise covariance matrix $\Phi_{\mathbf{v}}(f)$. In practice, this matrix is unknown and must be estimated from the observed data, making estimation errors unavoidable. Since the performance of the MVDR beamformer is highly sensitive to the accuracy of this estimate, obtaining a reliable and accurate estimate of $\Phi_{\mathbf{v}}(f)$ remains a critical challenge that must be addressed.

III. PROPOSED METHOD

The proposed framework estimates the NCM, $\Phi_{\mathbf{v}}(f)$, through a parametric formulation driven by a global low-dimensional SNR regression. Specifically, a lightweight neural network is first employed to estimate the input SNR, which is then used to construct the noise coherence matrix $\Gamma_{\mathbf{v}}(f)$. Since the coherence matrix corresponds to the covariance matrix normalized by the noise variance, and MVDR beamforming weights are scale-invariant, estimating $\Gamma_{\mathbf{v}}(f)$ is sufficient for implementation of MVDR.

A. Noise Coherence Matrix Estimation

For small and compact arrays, the noise variance at different sensors can be assumed to be identical, i.e., $\phi_{V_1}(f) = \phi_{V_2}(f) = \cdots = \phi_{V_M}(f) = \phi_V(f)$, where $\phi_{V_m}(f) = E[|V_m(f)|^2]$ for $m = 1, 2, \dots, M$. Similarly, the observed noisy signals at different sensors can also be assumed to be identical, i.e., $\phi_{Y_1}(f) = \phi_{Y_2}(f) = \cdots = \phi_{Y_M}(f) = \phi_Y(f)$, where $\phi_{Y_m}(f) = E[|Y_m(f)|^2] = \phi_X(f) + \phi_V(f)$ for $m = 1, 2, \dots, M$. Based on these conditions and with the assumption that speech and noise signals are mutually uncorrelated, we define the input SNR (iSNR) with respect to the reference microphone as

$$\text{iSNR}(f) = \frac{\phi_X(f)}{\phi_V(f)}. \quad (10)$$

The coherence matrix of $\mathbf{y}(f)$ can then be defined as

$$\Gamma_{\mathbf{y}}(f) = \frac{\Phi_{\mathbf{y}}(f)}{\phi_Y(f)}. \quad (11)$$

Substituting (6) into (11) yields [1]

$$\begin{aligned} \Gamma_{\mathbf{y}}(f) &= \frac{\phi_X(f)}{\phi_Y(f)} \Gamma_{\mathbf{x}}(f) + \frac{\phi_V(f)}{\phi_Y(f)} \Gamma_{\mathbf{v}}(f) \\ &= \frac{\text{iSNR}(f)}{1 + \text{iSNR}(f)} \Gamma_{\mathbf{x}}(f) + \frac{1}{1 + \text{iSNR}(f)} \Gamma_{\mathbf{v}}(f), \end{aligned} \quad (12)$$

where $\Gamma_{\mathbf{v}}(f) = \Phi_{\mathbf{v}}(f)/\phi_V(f)$ is the noise coherence matrix, and $\Gamma_{\mathbf{x}}(f) = \mathbf{d}_{\theta_s}(f)\mathbf{d}_{\theta_s}^H(f)$ is the desired signal coherence matrix.

From (12), the noise coherence matrix can be derived as

$$\begin{aligned} \Gamma_{\mathbf{v}}(f) &= [1 + \text{iSNR}(f)] \Gamma_{\mathbf{y}}(f) - \text{iSNR}(f) \Gamma_{\mathbf{x}}(f) \\ &= \Gamma_{\mathbf{y}}(f) + \text{iSNR}(f) [\Gamma_{\mathbf{y}}(f) - \Gamma_{\mathbf{x}}(f)]. \end{aligned} \quad (13)$$

This formulation is crucial as it shows that an accurate estimate of the iSNR allows for the direct computation of $\Gamma_{\mathbf{v}}(f)$ from the coherence matrices $\Gamma_{\mathbf{y}}(f)$, offering an alternative to conventional noise estimation techniques.

It has been established in the literature that SNR is closely related to the Wiener gain [32], [33], as shown by the following equation:

$$H_W(f) = \frac{\phi_X(f)}{\phi_Y(f)} = \frac{\text{iSNR}(f)}{1 + \text{iSNR}(f)}. \quad (14)$$

This relationship can be leveraged for robust and stable estimation of the noise coherence matrix, as the Wiener gain is bounded between 0 and 1, providing a straightforward means to control and stabilize the estimation process.

Substituting (14) into (12) leads to the following crucial relationship:

$$\Gamma_{\mathbf{y}}(f) = H_W(f)\mathbf{d}(f)\mathbf{d}^H(f) + [1 - H_W(f)] \Gamma_{\mathbf{v}}(f), \quad (15)$$

from which it can be deduced that

$$\hat{\Gamma}_{\mathbf{v}}(f) = \frac{\hat{\Gamma}_{\mathbf{y}}(f) - \hat{H}_W(f)\mathbf{d}(f)\mathbf{d}^H(f)}{1 - \hat{H}_W(f)}. \quad (16)$$

This relationship is essential for estimating the noise coherence

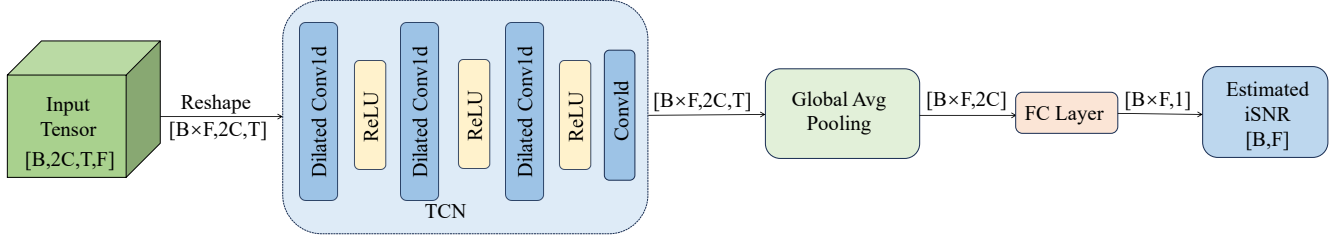


Fig. 1: Architecture of the proposed lightweight TCN for SNR estimation.

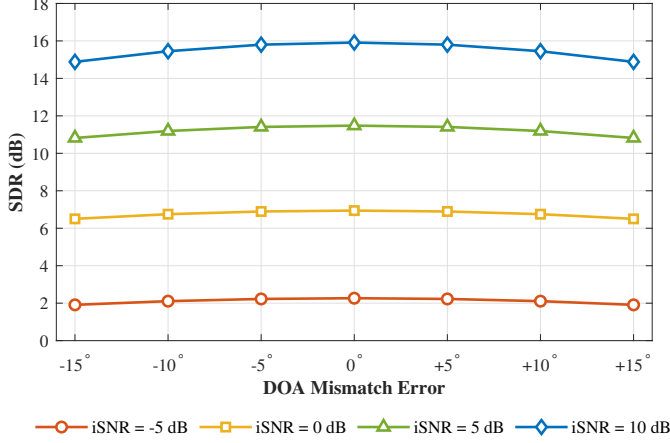


Fig. 2: SDR performance of the neural SNR estimation-assisted MVDR beamformer under different DOA mismatch errors and input iSNR conditions.

matrix once the Wiener gain and the coherence matrix estimates are obtained.

B. Lightweight Neural SNR Estimation

As discussed in the previous subsection, estimating the noise coherence matrix reduces to estimating the iSNR for each frequency bin. This estimated iSNR is subsequently used to calculate the Wiener gain, as shown in (14), which acts as the weighting factor in the reconstruction of the noise coherence matrix, as outlined in (15). By leveraging this approach, the complex problem of NCM estimation is transformed into a more manageable low-dimensional regression task.

To perform this regression, we utilize a lightweight Temporal Convolutional Network (TCN) with only 0.029M parameters, as shown in Fig. 1. The TCN is employed to estimate frequency-dependent SNR from multichannel observations [35], [36], demonstrating that the proposed SNR regression method offers strong performance with low computational complexity. While this particular architecture is used for illustration, the framework is not limited to this network design. The network takes multichannel STFT features as input and outputs a single SNR estimate for each frequency bin.

In detail, the real and imaginary parts of the complex STFT coefficients are concatenated along the channel dimension, forming an input tensor with the shape $[B, 2C, T, F]$, where B is the batch size, C is the number of channels, T is the time axis, and F is the frequency axis. The frequency dimension is processed independently, so the tensor is reshaped to $[B \times F, 2C, T]$ before passing it through the TCN.

The TCN consists of three dilated 1D convolutional layers, each with a kernel size of 3, a hidden dimension of 64, and dilation factors of 1, 2, and 4. ReLU activations are applied between layers,

followed by a 1×1 convolution to map the features to the output dimension.

After the temporal convolutions, global average pooling is applied along the temporal axis for each channel, producing a feature vector of length equal to the number of input channels. This operation aggregates long-term temporal dependencies while suppressing short-term fluctuations. The resulting vector is passed through a fully connected layer to generate a single SNR estimate for each frequency bin. Finally, the output is reshaped from $[B \times F, 1]$ to a tensor of shape $[B, 1, 1, F]$, which serves as the frequency-dependent SNR estimate required for constructing the noise coherence matrix $\Gamma_v(f)$.

C. MVDR Beamforming Implementation

With the estimated noise coherence matrix, we implement the MVDR beamformer in the STFT domain. For each frequency bin, the MVDR beamformer is applied as follows:

$$\mathbf{h}_{\text{MVDR}}(f) = \frac{\hat{\Gamma}_v^{-1}(f) \mathbf{d}_{\theta_s}(f)}{\mathbf{d}_{\theta_s}^H(f) \hat{\Gamma}_v^{-1}(f) \mathbf{d}_{\theta_s}(f)}. \quad (17)$$

IV. EXPERIMENTS AND RESULTS

This section assesses the performance of the MVDR beamformer using the proposed frequency-domain neural SNR estimation approach.

A. Experimental Setup

Multichannel speech signals were simulated using room impulse responses (RIRs) generated with randomly selected room parameters [37]. Room dimensions were uniformly sampled, with length and width drawn from [4, 10] m, and height from [3, 4] m. Reverberation times (T_{60}) ranged between 100 and 400 ms, ensuring a variety of low reverberant environments.

A uniform linear array (ULA) with 8 microphones and a 0.02 m inter-element spacing was placed at the center of the room at a height of 1.5 m. The target speech source was positioned at $\theta_s = 90^\circ$, with the distance randomly selected from [1.0, 2.0] m. The generated RIRs were broken down into direct-path, early-reflection, and late-reflection components to facilitate beamforming evaluation. To simulate multichannel speech propagation with multi-path effects, the VCTK corpus [38] was used for target speech, while the DEMAND dataset [39] provided interference and environmental noise. Diffuse noise was simulated based on the array geometry and scaled to various input SNR levels. All speech signals were resampled to 16 kHz before data synthesis to ensure uniformity in RIR convolution and beamforming processes. Noise sources included street, crowd, and indoor environments, which helped improve the model's generalization ability for diverse interference scenarios.

TABLE I: Performance of the MVDR-Oracle, MVDR-iSNR-Oracle, MVDR-iSNR-IMCRA, and MVDR-iSNR-NN methods under low-reverberation conditions ($T_{60} < 200$ ms).

Method	iSNR = -5 dB		iSNR = 0 dB		iSNR = 5 dB		iSNR = 10 dB	
	STOI	SDR	STOI	SDR	STOI	SDR	STOI	SDR
Noisy	0.69	-4.40	0.78	0.45	0.87	5.46	0.92	10.35
MVDR-Oracle	0.85	5.54	0.92	10.48	0.96	15.48	0.98	20.33
MVDR-iSNR-Oracle	0.83	4.29	0.90	8.28	0.94	12.20	0.96	15.96
MVDR-iSNR-IMCRA	0.67	-0.61	0.79	4.69	0.90	10.13	0.95	15.33
MVDR-iSNR-NN	0.78	2.27	0.87	6.94	0.93	11.48	0.96	15.91

TABLE II: Performance of the MVDR-Oracle, MVDR-iSNR-Oracle, MVDR-iSNR-IMCRA, and MVDR-iSNR-NN methods under high-reverberation conditions ($T_{60} > 300$ ms).

Method	iSNR = -5 dB		iSNR = 0 dB		iSNR = 5 dB		iSNR = 10 dB	
	STOI	SDR	STOI	SDR	STOI	SDR	STOI	SDR
Noisy	0.65	-2.93	0.72	1.37	0.77	4.55	0.81	6.86
MVDR-Oracle	0.82	5.40	0.88	8.87	0.91	11.04	0.93	12.41
MVDR-iSNR-Oracle	0.80	4.28	0.85	7.09	0.88	8.64	0.90	9.48
MVDR-iSNR-IMCRA	0.63	0.54	0.72	4.31	0.79	6.70	0.84	8.25
MVDR-iSNR-NN	0.74	2.77	0.81	6.16	0.86	8.31	0.88	9.75

The STFT was computed using a frame size of 512 samples with a Hanning window and a 50% overlap. The real and imaginary parts of the STFT were concatenated to form a 4D tensor of shape $[B, 2C, T, F]$, which was then fed into the proposed TCN-based iSNR estimator. The network outputs frequency-dependent iSNR estimates, which are subsequently used to construct the noise coherence matrix for MVDR beamforming.

The experiment consists of the following four configurations.

1) MVDR-Oracle: In this configuration, the ideal speech and noise covariance matrices, $\Phi_x(f)$ and $\Phi_v(f)$, are directly used to compute the MVDR beamforming weights $\mathbf{h}_{\text{MVDR}}$ [34]. The resulting MVDR output serves as the theoretical upper bound on performance under the assumption of ideal spatial covariance information.

2) MVDR-iSNR-Oracle: Here, the noise coherence matrix $\Gamma_v(f)$ is computed using ideal frequency-wise iSNR values to explore whether $\Gamma_v(f)$ can be constructed purely from iSNR information.

3) MVDR-iSNR-NN: In this configuration, the frequency-domain iSNR is estimated by a neural network. The estimated iSNR values are then used to derive $\Gamma_v(f)$, which is employed to compute the MVDR beamforming weights for speech enhancement.

4) MVDR-iSNR-IMCRA: A similar workflow is followed, where a priori SNR is first estimated using the IMCRA method [40]. This estimated SNR is then used to construct $\Gamma_v(f)$ for stabilized MVDR beamforming weights. This configuration serves as a baseline for performance comparison.

B. Results and Analysis

We first evaluate all configurations under low-reverberation conditions ($T_{60} < 200$ ms) at fixed input iSNR levels of 10 dB, 5 dB, 0 dB, and -5 dB. The performance results under these conditions are summarized in Table I.

It is seen that MVDR-Oracle achieves the best performance and serves as the theoretical upper bound. MVDR-iSNR-Oracle closely approaches this upper bound, demonstrating that constructing the NCM using the ideal frequency-wise iSNR yields highly competitive performance for MVDR beamforming.

Both MVDR-iSNR-NN and MVDR-iSNR-IMCRA estimate the frequency-dependent iSNR to reconstruct the noise covariance

matrix. Compared to the MVDR-iSNR-IMCRA approach, the proposed MVDR-iSNR-NN, which leverages neural network-based iSNR estimation, consistently delivers superior performance across all tested SNR levels. As the input iSNR decreases, the performance of all methods gradually degrades. However, the relative performance ranking among the methods remains consistent.

Results for stronger reverberation conditions ($T_{60} > 300$ ms) are presented in Table II. As expected, all methods perform worse under stronger reverberation, but their relative ranking remains unchanged.

In practical scenarios, the direction of arrival (DOA) of the target source is typically unknown and must be estimated in order to construct the steering vector. As shown in (16), the reconstruction of the noise coherence matrix depends directly on the steering vector. Therefore, errors in DOA estimation lead to mismatches in the steering vector, which in turn cause inaccuracies in the estimated noise coherence matrix, ultimately degrading the performance of MVDR beamforming.

To assess the robustness of the proposed method to steering vector mismatch, DOA estimation errors ranging from -15° to $+15^\circ$ were introduced in the steering vector construction. Figure 2 illustrates the SDR performance of the proposed SNR-estimated MVDR method under varying DOA mismatch conditions. The performance remains stable with only minimal degradation across all input SNR levels, even when the DOA error reaches $\pm 15^\circ$.

These results demonstrate that the proposed approach is highly robust to moderate steering vector mismatch under the considered conditions.

V. CONCLUSIONS

This paper introduces an efficient implementation of the MVDR beamformer, where the NCM is estimated via frequency-wise iSNR regression. A lightweight TCN is used to estimate the iSNR from multichannel spectral features. The estimated iSNR is then integrated into the coherence matrix reconstruction for computing the MVDR beamforming weights. Experimental results demonstrate that the approach delivers robust and consistent speech enhancement performance with low computational overhead, while also showing resilience to steering vector mismatch, making it highly suitable for practical array processing applications.

REFERENCES

- [1] J. Benesty, G. Huang, J. Chen, and N. Pan, *Microphone Arrays*, vol. 22. Berlin, Germany: Springer-Verlag, 2023.
- [2] J. Benesty, I. Cohen, and J. Chen, *Fundamentals of Signal Enhancement and Array Signal Processing*. Singapore: Wiley-IEEE Press., 2018.
- [3] M. Brandstein and D. Ward, *Microphone Arrays: Signal Processing Techniques and Applications*. Springer, 2001.
- [4] G. Huang, J. R. Jensen, J. Chen, J. Benesty, M. G. Christensen, A. Sugiyama, G. Elko, and T. Gaensler, "Advances in microphone array processing and multichannel speech enhancement," in *Proc. IEEE ICASSP*, pp. 1–5, 2025.
- [5] G. W. Elko and J. Meyer, "Microphone arrays," in *Springer Handbook of Speech Processing* (J. Benesty, M. M. Sondhi, and Y. Huang, eds.), ch. 48, pp. 1021–1041, Berlin, Germany: Springer-Verlag, 2008.
- [6] J. Benesty, J. Chen, and Y. Huang, *Microphone Array Signal Processing*. Berlin, Germany: Springer-Verlag, 2008.
- [7] G. Huang, J. Benesty, and J. Chen, "On the design of frequency-invariant beamformers with uniform circular microphone arrays," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 25, no. 5, pp. 1140–1153, 2017.
- [8] G. Huang, J. Benesty, and J. Chen, "Fundamental approaches to robust differential beamforming with high directivity factors," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 30, pp. 3074–3088, 2022.
- [9] S. Gannot, E. Vincent, S. Markovich-Golan, A. Ozerov, S. Gannot, E. Vincent, S. Markovich-Golan, and A. Ozerov, "A consolidated perspective on multi-microphone speech enhancement and source separation," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 25, no. 4, pp. 692–730, 2017.
- [10] Z. Zhang, Y. Xu, M. Yu, S.-X. Zhang, L. Chen, and D. Yu, "ADL-MVDR: all deep learning MVDR beamformer for target speech separation," in *Proc. IEEE ICASSP*, pp. 6089–6093, 2021.
- [11] Y. Fu, M. Ge, H. Wang, N. Li, H. Yin, L. Wang, G. Zhang, J. Dang, C. Deng, and F. Wang, "Locate and beamform: Two-dimensional locating all-neural beamformer for multi-channel speech separation," *arXiv preprint arXiv:2305.10821*, 2023.
- [12] V. M. Tavakoli, J. R. Jensen, M. G. Christensen, and J. Benesty, "Pseudo-coherence-based MVDR beamformer for speech enhancement with ad hoc microphone arrays," in *Proc. IEEE ICASSP*, pp. 2659–2663, 2015.
- [13] W. Liu, J. Benesty, G. Huang, and J. Chen, "Beamforming in the short-time fourier transform domain via dimensionality reduction," *IEEE Trans. Audio, Speech, Lang. Process.*, 2025.
- [14] J. Heymann, L. Drude, and R. Haeb-Umbach, "Neural network based spectral mask estimation for acoustic beamforming," in *Proc. IEEE ICASSP*, pp. 196–200, 2016.
- [15] C. Boeddeker, H. Erdogan, T. Yoshioka, and R. Haeb-Umbach, "Exploring practical aspects of neural mask-based beamforming for far-field speech recognition," in *Proc. IEEE ICASSP*, pp. 6697–6701, 2018.
- [16] C. Pan, J. Chen, et al., "Performance study of the mvdr beamformer as a function of the source incidence angle," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 22, no. 1, pp. 67–79, 2014.
- [17] E. A. P. Habets, J. Benesty, I. Cohen, S. Gannot, and J. Dmochowski, "New insights into the MVDR beamformer in room acoustics," *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 18, no. 1, p. 158, 2010.
- [18] W. Liu, G. Huang, X. Liu, J. Jin, J. Chen, and J. Benesty, "Microphone array beamforming for speech enhancement based on dynamic mode decomposition," in *Proc. IEEE ICASSP*, 2025.
- [19] J. Benesty, J. Chen, and E. A. Habets, *Speech Enhancement in the STFT Domain*. Berlin, Germany: Springer-Verlag, 2011.
- [20] S. Araki, N. Ono, K. Kinoshita, and M. Delcroix, "Meeting recognition with asynchronous distributed microphone array using block-wise refinement of mask-based mvdr beamformer," in *Proc. IEEE ICASSP*, pp. 5694–5698, 2018.
- [21] M. Souden, E. Vincent, and M. Moonen, "On optimal frequency-domain multichannel linear filters for noise reduction," *IEEE Trans. Speech, Audio Process.*, vol. 19, pp. 1294–1306, 2011.
- [22] F. Zhang, C. Pan, J. Benesty, and J. Chen, "Directional gain based noise covariance matrix estimation for mvdr beamforming," in *Proc. IEEE ICASSP*, pp. 511–515, 2024.
- [23] L. Drude and R. Haeb-Umbach, "Integration of neural networks and probabilistic spatial models for acoustic blind source separation," *IEEE J. Sel. Topics Signal Process.*, vol. 13, no. 4, pp. 815–826, 2019.
- [24] T. Higuchi, N. Ito, S. Araki, T. Yoshioka, M. Delcroix, and T. Nakatani, "Online MVDR beamformer based on complex gaussian mixture model with spatial prior for noise robust ASR," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 25, no. 4, pp. 780–793, 2017.
- [25] H. Erdogan, J. R. Hershey, S. Watanabe, M. I. Mandel, and J. Le Roux, "Improved mvdr beamforming using single-channel mask prediction networks," in *Interspeech*, pp. 1981–1985, 2016.
- [26] C.-H. Lee, K. Patel, C. Yang, Y. Shen, and H. Jin, "An mvdr-embedded u-net beamformer for effective and robust multichannel speech enhancement," in *Proc. IEEE ICASSP*, pp. 8541–8545, 2024.
- [27] H. V. Trees, *Optimum Array Processing: Part IV of Detection, Estimation, and Modulation theory*. New York, John Wiley Sons, Inc, 2002.
- [28] B. D. Van Veen and K. M. Buckley, "Beamforming: A versatile approach to spatial filtering," *IEEE assp magazine*, vol. 5, no. 2, pp. 4–24, 1988.
- [29] T. Higuchi, N. Ito, T. Yoshioka, and T. Nakatani, "Robust mvdr beamforming using time-frequency masks for online/offline asr in noise," in *Proc. IEEE ICASSP*, pp. 5210–5214, 2016.
- [30] S. S. Haykin, *Adaptive Filter Theory*. Pearson Education India, 2002.
- [31] J. Capon, "High-resolution frequency-wavenumber spectrum analysis," *Proc. IEEE*, vol. 57, pp. 1408–1418, Aug. 1969.
- [32] Y. Ephraim and D. Malah, "Speech enhancement using a minimum-mean square error short-time spectral amplitude estimator," *IEEE Trans. Speech, Audio Process.*, vol. 32, no. 6, pp. 1109–1121, 2003.
- [33] P. Loizou, *Speech Enhancement: Theory and Practice*. Boca Raton, FL, CRC Press, 2007.
- [34] J. Benesty, J. Chen, and Y. Huang, *Microphone Array Signal Processing*. Springer, 2017.
- [35] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. arxiv," *arXiv preprint arXiv:1803.01271*, vol. 10, 2018.
- [36] Y. Luo and N. Mesgarani, "Conv-tasnet: Surpassing ideal time-frequency magnitude masking for speech separation," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 27, no. 8, pp. 1256–1266, 2019.
- [37] E. A. Habets, "Room impulse response generator," *Technische Universiteit Eindhoven, Tech. Rep.*, vol. 2, no. 2.4, p. 1, 2006.
- [38] J. Yamagishi, C. Veaux, and K. MacDonald, "CSTR VCTK corpus: English multi-speaker corpus for CSTR voice cloning toolkit (version 0.92)," *Univ. Edinburgh, CSTR*, 2019.
- [39] J. Thiemann, N. Ito, and E. Vincent, "Demand: a collection of multi-channel recordings of acoustic noise in diverse environments," June 2013.
- [40] I. Cohen, "Noise spectrum estimation in adverse environments: Improved minima controlled recursive averaging," *IEEE Trans. Speech, Audio Process.*, vol. 11, no. 5, pp. 466–475, 2003.